

Java String Functions — Cheat Sheet

Common **java.lang.String** methods with signatures, short descriptions, examples, and results. Strings are immutable: methods return a new String unless noted.

Method	Description	Example	Result
<code>length()</code>	Returns number of characters	<code>s = "Hello"; s.length()</code>	5
<code>isEmpty()</code>	true if length is 0	<code>"".isEmpty()</code>	true
<code>isBlank()</code>	true if empty or only whitespace (Java 11+)	<code>" ".isBlank()</code>	true
<code>charAt(int index)</code>	Char at given index (0-based)	<code>"Java".charAt(2)</code>	'v'
<code>codePointAt(int index)</code>	Unicode code point at index	<code>"A".codePointAt(0)</code>	65
<code>codePointBefore(int index)</code>	Code point before index	<code>"AB".codePointBefore(1)</code>	65
<code>codePointCount(int b, int e)</code>	Code points in range [b, e)	<code>"Hi".codePointCount(0, 2)</code>	2
<code>toCharArray()</code>	Converts string to char array	<code>"Hi".toCharArray()</code>	['H', 'i']
<code>getChars(srcBegin, srcEnd, dst, dstBegin)</code>	Copies chars into array	See docs; fills dst[]	—
<code>indexOf(String str)</code>	First index of substring (-1 if absent)	<code>"Hello".indexOf("l")</code>	2
<code>indexOf(String str, int from)</code>	First index from given offset	<code>"Hello".indexOf("l", 3)</code>	3
<code>indexOf(int ch)</code>	First index of character	<code>"Java".indexOf('a')</code>	1
<code>lastIndexOf(String str)</code>	Last index of substring	<code>"Hello".lastIndexOf("l")</code>	3
<code>lastIndexOf(String str, int from)</code>	Last index searching backward from	<code>"Hello".lastIndexOf("l", 2)</code>	2
<code>contains(CharSequence s)</code>	true if substring found	<code>"Java".contains("av")</code>	true
<code>startsWith(String prefix)</code>	true if starts with prefix	<code>"Hello".startsWith("He")</code>	true
<code>startsWith(String prefix, int toffset)</code>	true if substring at offset starts with	<code>"Hello".startsWith("ll", 2)</code>	true
<code>endsWith(String suffix)</code>	true if ends with suffix	<code>"Hello".endsWith("lo")</code>	true
<code>matches(String regex)</code>	true if entire string matches regex	<code>"123".matches("\\d+")</code>	true
<code>regionMatches(...)</code>	Compares a region with another string	<code>"Hello".regionMatches(0, "He", 0, 2)</code>	true
<code>equals(Object obj)</code>	Case-sensitive equality	<code>"Hi".equals("Hi")</code>	true
<code>equalsIgnoreCase(String another)</code>	Case-insensitive equality	<code>"Hi".equalsIgnoreCase("hi")</code>	true
<code>compareTo(String another)</code>	Lexicographic compare (0 = equal)	<code>"A".compareTo("B")</code>	-1
<code>compareToIgnoreCase(String another)</code>	Case-insensitive compare	<code>"a".compareToIgnoreCase("A")</code>	0
<code>contentEquals(CharSequence cs)</code>	Content equals CharSequence/StringBuffer	<code>"ab".contentEquals("ab")</code>	true

Method	Description	Example	Result
<code>substring(int begin)</code>	Substring from begin to end	<code>"Hello".substring(2)</code>	<code>"llo"</code>
<code>substring(int begin, int end)</code>	Substring [begin, end)	<code>"Hello".substring(1, 4)</code>	<code>"ell"</code>
<code>subSequence(int begin, int end)</code>	CharSequence for [begin, end)	<code>"Hello".subSequence(1, 4)</code>	<code>"ell"</code>
<code>concat(String str)</code>	Concatenates strings	<code>"Hello".concat("!")</code>	<code>"Hello!"</code>
<code>replace(char old, char new)</code>	Replace all matching chars	<code>"aabb".replace('a', 'x')</code>	<code>"xxbb"</code>
<code>replace(CharSequence t, CharSequence r)</code>	Replace all matching sequences	<code>"ab ab".replace("ab", "X")</code>	<code>"X X"</code>
<code>replaceAll(String regex, String repl)</code>	Replace all regex matches	<code>"a1b2".replaceAll("\\d", "")</code>	<code>"ab"</code>
<code>replaceFirst(String regex, String repl)</code>	Replace first regex match	<code>"a1b2".replaceFirst("\\d", "")</code>	<code>"ab2"</code>
<code>toLowerCase()</code>	Lowercase (default locale)	<code>"JAVA".toLowerCase()</code>	<code>"java"</code>
<code>toUpperCase()</code>	Uppercase (default locale)	<code>"java".toUpperCase()</code>	<code>"JAVA"</code>
<code>trim()</code>	Strip leading/trailing whitespace (<= U+0020)	<code>" Hi ".trim()</code>	<code>"Hi"</code>
<code>strip()</code>	Strip leading/trailing Unicode whitespace (Java 11+)	<code>" Hi ".strip()</code>	<code>"Hi"</code>
<code>stripLeading()</code>	Strip leading whitespace (Java 11+)	<code>" Hi".stripLeading()</code>	<code>"Hi"</code>
<code>stripTrailing()</code>	Strip trailing whitespace (Java 11+)	<code>"Hi ".stripTrailing()</code>	<code>"Hi"</code>
<code>intern()</code>	Returns canonical pool string	<code>"Hi".intern()</code>	pooled ref
<code>repeat(int count)</code>	Repeat string count times (Java 11+)	<code>"ab".repeat(3)</code>	<code>"ababab"</code>
<code>indent(int n)</code>	Adjust indentation (Java 12+)	<code>"a\nb".indent(2)</code>	<code>" a\n b\n"</code>
<code>stripIndent()</code>	Strip incidental indentation (Java 15+)	<code>"""\n a\n b\n""".stripIndent()</code>	<code>"a\nb"</code>
<code>translateEscapes()</code>	Translate escape sequences (Java 15+)	<code>"a\\nb".translateEscapes()</code>	<code>"a\nb"</code>
<code>transform(Function f)</code>	Apply function; return result (Java 12+)	<code>"hi".transform(String::toUpperCase)</code>	<code>"HI"</code>
<code>split(String regex)</code>	Split around regex matches	<code>"a,b,c".split(",")</code>	<code>["a", "b", "c"]</code>
<code>split(String regex, int limit)</code>	Split with limit	<code>"a,b,c".split(",", 2)</code>	<code>["a", "b,c"]</code>
<code>join(CharSequence delim, CharSequence... elems) [static]</code>	Join elements with delimiter	<code>String.join("-", "a", "b")</code>	<code>"a-b"</code>
<code>lines()</code>	Stream of lines (Java 11+)	<code>"a\nb".lines().toList()</code>	<code>["a", "b"]</code>
<code>format(String format, Object... args) [static]</code>	Formatted string	<code>String.format("%s %d", "Hi", 2)</code>	<code>"Hi 2"</code>
<code>formatted(Object... args)</code>	Instance format (Java 15+)	<code>"%s %d".formatted("Hi", 2)</code>	<code>"Hi 2"</code>
<code>valueOf(...) [static]</code>	Converts value to String	<code>String.valueOf(42)</code>	<code>"42"</code>
<code>copyValueOf(char[] data) [static]</code>	String from char array	<code>String.copyValueOf(new char[]{'a'})</code>	<code>"a"</code>

Method	Description	Example	Result
<code>getBytes()</code>	Encode to bytes (platform default charset)	<code>"Hi".getBytes()</code>	<code>byte[]</code>
<code>getBytes(Charset charset)</code>	Encode with charset	<code>"Hi".getBytes(StandardCharsets.UTF_8)</code>	<code>byte[]</code>
<code>hashCode()</code>	Hash code of string content	<code>"Hi".hashCode()</code>	<code>int</code>
<code>toString()</code>	Returns this string	<code>"Hi".toString()</code>	<code>"Hi"</code>
<code>chars()</code>	IntStream of char values	<code>"AB".chars().toArray()</code>	<code>[65, 66]</code>
<code>codePoints()</code>	IntStream of code points	<code>"A".codePoints().toArray()</code>	<code>[65]</code>
<code>describeConstable()</code>	Optional describing this string (Java 12+)	<code>"Hi".describeConstable()</code>	<code>Optional[Hi]</code>
<code>resolveConstantDesc(Lookup)</code>	Resolves constant (Java 12+)	See <code>Constable API</code>	<code>String</code>

Notes: (1) `IndexOutOfBoundsException` if index is invalid. (2) `NullPointerException` if a required argument is null (e.g. `equalsIgnoreCase(null)`). (3) Prefer `strip()` over `trim()` for Unicode whitespace. (4) Use `Objects.equals(a, b)` for null-safe equality. (5) For building many concatenations in a loop, use `StringBuilder`.

Total methods covered: **61** | Reference: Oracle Java SE String API (Java 11–21 features noted where relevant).